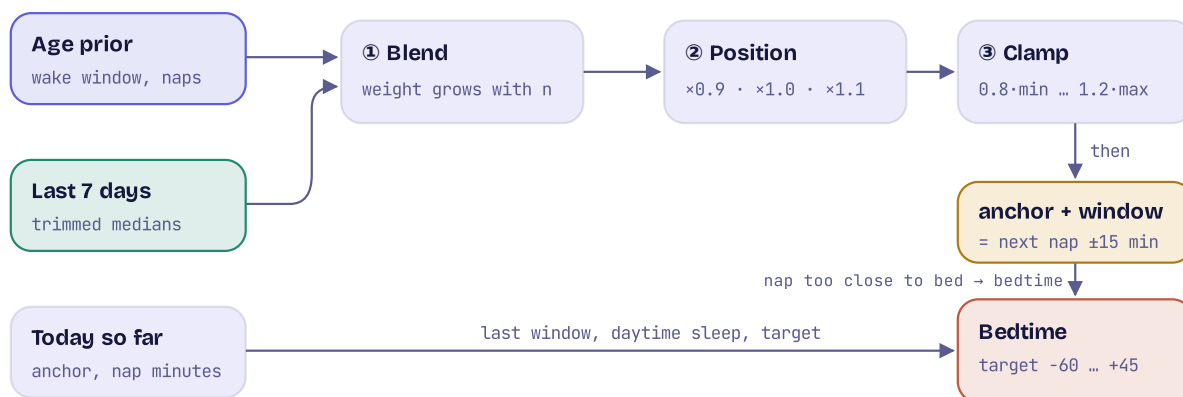


How Nuggi guesses the next nap

No machine learning, no server call. Start from what's typical for the baby's age, lean more on the baby's own logged rhythm the more of it there is, then count forward from the last time the baby woke up. The same inputs always give the same answer, and every step writes a line into the “Warum?” (“why?”) panel on the Heute screen.

OVERVIEW

The whole pipeline on one page



Start from what's typical for the age

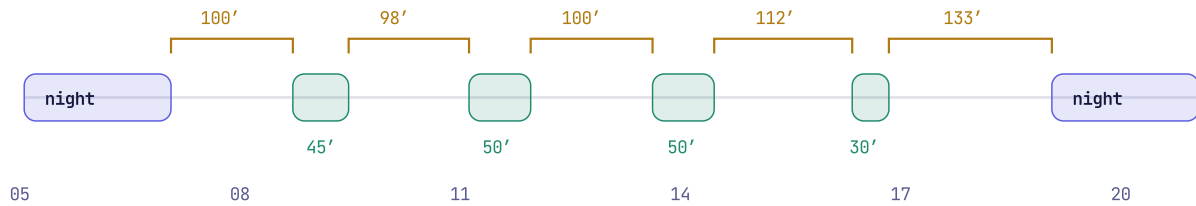
The baby's age in completed weeks picks one row. The midpoint of the wake window range is the starting guess; nap count and length feed into bedtime and the nap duration.

BRACKET	FROM WK	WAKE WINDOW	MIDPOINT	NAPS	NAP LENGTH	CLAMP RANGE
0-4 weeks	0	35-60'	47.5'	4-6	40'	28-72'
5-12 weeks	5	60-90'	75'	4-5	45'	48-108'
3-4 months	13	75-120'	97.5'	3-4	45'	60-144'
5-6 months	22	120-150'	135'	3	60'	96-180'
7-9 months	31	150-210'	180'	2-3	75'	120-252'
10-12 months	44	180-240'	210'	2	75'	144-288'
13-18 months	57	240-330'	285'	1-2	90'	192-396'
19+ months	83	300-360'	330'	1	100'	240-432'

Clamp range = 0.8 × min ... 1.2 × max. The prediction can never leave it, however odd the logged data looks.

Measure the baby's own rhythm

Finished sleeps from the last 7 days are sorted by start time. The gap between one sleep ending and the next starting is one *wake window*. Night wakings don't count, and neither do implausible gaps.



Counts as a wake window

A gap of 10 min to 8 h between two logged sleeps. A night→night gap that ends inside the night window (default 19:00–07:00) is a night waking and is skipped.

Counts as a nap length

A sleep tagged `nap` lasting 5–240 min.

Trimmed median

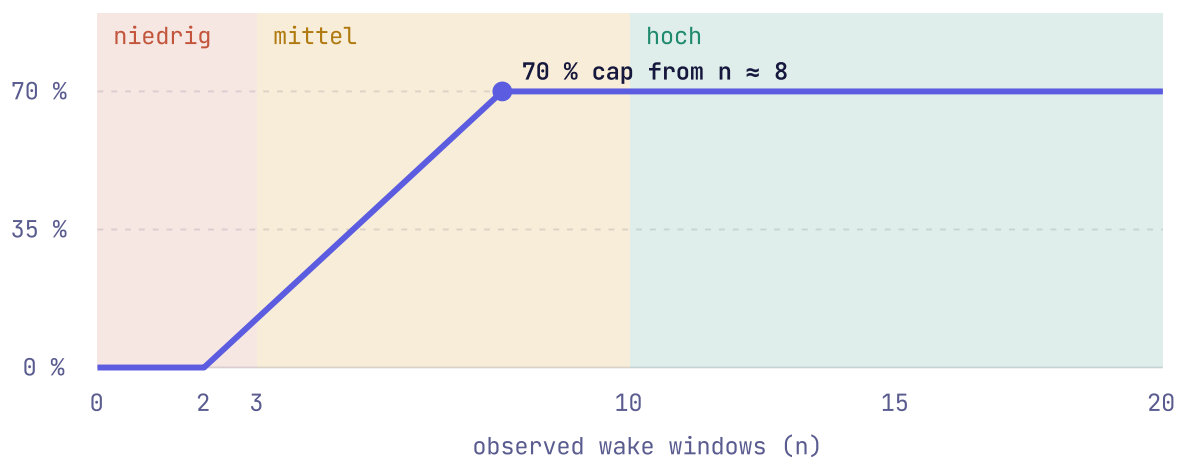
Sort, drop the top and bottom 10 % (rounded down), take the median. One skipped nap or a 20-minute catnap can't pull the result much.

STEP 3 · BLENDWEIGHT()

Trust the baby more as the data grows

With n observed wake windows, the observed median gets weight w and the age prior gets $1 - w$. It starts after 2 observations and tops out at 70 %, so the age norm always keeps at least 30 % of the say.

```
w = min(0.7, max(0, (n - 2) / 8))
window = (1 - w) · prior_mid · position + w · observed_median
window = clamp(window, 0.8 · min, 1.2 · max) // rounded to whole minutes
```



The same n sets the confidence label shown on the ring: under 3 is **niedrig**, 3–9 **mittel**, 10+ **hoch**. Nap length uses the identical blend on its own count, clamped to $0.6\text{--}1.6 \times$ the prior nap length.

STEP 4 · POSITIONFACTOR()

First window shorter, last one longer

Babies usually manage less awake time right after the night and more before bed. The prior part of the blend is scaled by where in the day the upcoming gap sits, counted by naps already done today (a nap in progress counts as done).



Count forward from the right moment

The next nap is **anchor + wake window**, shown as a window of ± 15 min around it. Which anchor depends on the current state:

asleep • nap

Predicted wake-up

Nap start + nap length (at least now + 5 min). The following window counts from there.

asleep • night

Night end

Wake-up is the configured night end (07:00 by default); tomorrow's bedtime is shown.

awake

Last sleep ended

Used if it ended within the last 16 hours.

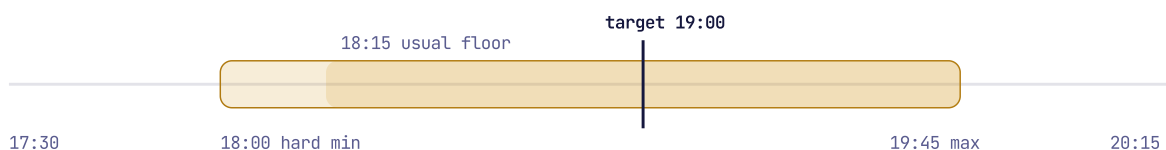
no data

Assumed wake-up

Night end, or “now” if it's still before that. The reasoning says so explicitly.

Bedtime bends toward the target, but only so far

```
bedtime = max(lastNapEnd + lastWindow, target - 45)
bedtime = clamp(bedtime, target - 60, target + 45)
if daytime naps today > expected + 30 min           → bedtime +15
if daytime naps today < expected - 45 min (after target - 3 h) → bedtime -15
if nextNap > bedtime - napLength / 2                 → no more naps: next sleep
is bedtime
```



The light band is where the formula normally lands; the darker edge down to target – 60 is only reachable via the “little daytime sleep” adjustment. Expected daytime sleep = average naps $\times$ prior nap length ($3.5 \times 45 = 158$ min at 3–4 months).

Simulate a day

This runs the same formulas as `predict()` : wake at 07:00, bedtime target 19:00, every nap taking exactly the predicted length. It leaves out the ± 15 min daytime-sleep nudge. Starts with a 4-month-old who has 12 logged wake windows.

AGE BRACKET

3–4 months

OBSERVED WINDOWS (N)

12

OBSERVED MEDIAN

110 min

OBSERVED NAP LENGTH

45 min

OBSERVED WEIGHT

70 %

prior keeps 30 %

FIRST WINDOW

103 min

 $\times 0.9$ on the prior

NAP LENGTH

45 min

prior 45 min

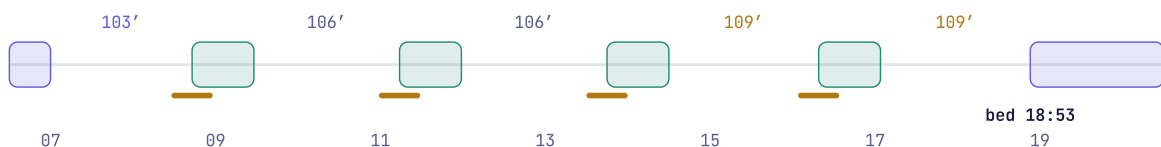
BEDTIME

18:53

after 4 naps

CONFIDENCE

HOCH

from $n = 12$ 

1. Age 3–4 months: typical wake window 75–120 min, midpoint 97.5.
2. Blend 70 % observed (110 min) + 30 % prior, clamped to 60–144 min.
3. 07:00 + 103 min ($\times 0.9$) → nap 1 at 08:43, window 08:28–08:58, ends 09:28.
4. 09:28 + 106 min ($\times 1.0$) → nap 2 at 11:14, window 10:59–11:29, ends 11:59.
5. 11:59 + 106 min ($\times 1.0$) → nap 3 at 13:45, window 13:30–14:00, ends 14:30.
6. 14:30 + 109 min ($\times 1.1$) → nap 4 at 16:19, window 16:04–16:34, ends 17:04.
7. 17:04 + 109 min would be 18:53, later than bedtime – 23 min → next sleep is the night at 18:53.

BONUS

Feeds use a plain interval

Next feed = last feed start + interval. The interval is the family's setting, otherwise by age: 150 min up to 4 weeks, 180 up to 12, 210 up to 30, then 240. The push reminders cron reads the same `Prediction` object, so the ring and the notifications never disagree.

Source: `src/lib/sleep/predict.ts`, `src/lib/sleep/priors.ts`, tests in `predict.test.ts`. The simulator is a JavaScript port for illustration; the app's own output wins where they differ.